

🔥 LiquidAI / **LFM2.5-230M** 📄

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 Text Generation
  Transformers
  Safetensors
  10 languages
 lfm2 liquid lfm2.5

edge conversational
  Eval Results
  arxiv:2511.23404
  License: lfm1.0



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 **Model card**
 Files
  xet
  Community 11



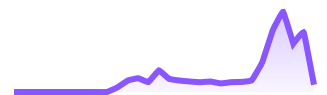
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LFM2.5-230M

LFM2.5 is a family of hybrid models designed for **on-device deployment**. It builds on the LFM2 architecture with extended pre-training and reinforcement learning.

- **Our most compact model yet:** 230M parameters that punch above their weight, bringing real capability to the tightest memory and compute budgets.
- **Fast edge inference:** Best throughput from low-cost CPUs to production GPUs, running at 213 tok/s decode speed on Galaxy S25 Ultra and 42 tok/s on a Raspberry Pi 5.
- **Built for agentic tasks:** Distilled from LFM2.5-350M and refined with multi-stage

Downloads last month
21,935



 **Safetensors** ⓘ

Model size 0.2B params

Tensor type BF16  Chat template

↗ Files info

⚡ **Inference Providers** **NEW**

 Text Generation

This model isn't deployed by any Inference Provider.

 Ask for provider support

 **Model tree for LiquidAI/LFM2.5-2...**

Base model LiquidAI/LFM2.5-230M-Ba...

 **Finetuned** (2) [this model](#)

 **Adapters** 2 models

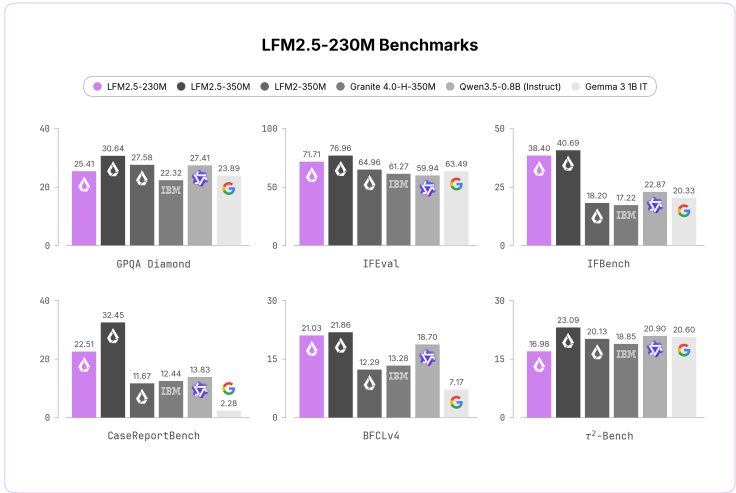
 **Finetunes** 15 models

 **Quantizations** ⚡ 📄 🙌 15 models

 **Spaces using LiquidAI/LFM2.5-...** 4

reinforcement learning, making it well-suited for tool use and data extraction.

Find more information about LFM2.5-230M in our [blog post](#).



Model Details

Model	Parameters	Description
LFM2.5-230M-Base	230M	Pre-trained base model for fine-tuning
LFM2.5-230M	230M	General-purpose instruction-tuned model

LFM2.5-230M is a general-purpose text-only model with the following features:

- **Number of parameters:** 230M
- **Number of layers:** 14 (8 double-gated LIV convolution blocks + 6 GQA blocks)
- **Training budget:** 19T tokens
- **Context length:** 32,768 tokens
- **Vocabulary size:** 65,536

- [webml-community/lfm2-webgpu...](#)
- [choltha/lfm2-webgpu-kernels](#)
- [Ramtinoss/lfm2-webgpu-kernels](#)
- [Ckimd/first_test](#)

Collection including LiquidAI/LFM...

LFM2.5 Collection

Collection ... • 14 items • Up... • Δ 168

Paper for LiquidAI/LFM2.5-230M

LFM2 Technical Report

Paper • 2511.23404 • Publi... • Δ 65

Evaluation results

- Idavidrein/gpqa · Diamond Δ 25.41
- TIGER-Lab/MMLU-Pro · Mml... Δ 20.25

- **Knowledge cutoff:** Mid-2024
- **Languages:** English, Arabic, Chinese, French, German, Italian, Japanese, Korean, Portuguese, Spanish
- **Generation parameters:**
 - temperature: 0.1
 - top_k: 50
 - repetition_penalty: 1.05

Model	Description
LFM2.5-230M	Original model checkpoint in native format. Best for fine-tuning or inference with Transformers, vLLM, and SGLang.
LFM2.5-230M-GGUF	Quantized format for llama.cpp and compatible tools. Optimized for edge inference and local deployment.
LFM2.5-230M-ONNX	ONNX Runtime format for cross-platform deployment.
LFM2.5-230M-MLX	MLX format for Apple Silicon. Optimized for fast inference on Mac devices.

We recommend using it for data extraction and lightweight on-device agentic pipelines. It is not recommended for reasoning-heavy workloads such as advanced math, code generation, or creative writing.

Chat Template

LFM2.5 uses a ChatML-like format. See the [Chat Template documentation](#) for details. Example:

```
<|startoftext|><|im_start|>system
You are a helpful assistant trained by Liquid
<|im_start|>user
What is C. elegans?<|im_end|>
<|im_start|>assistant
```

You can use `tokenizer.apply_chat_template()` to format your messages automatically.

Tool Use

LFM2.5 supports function calling in four steps:

1. **Function definition:** Provide the list of tools as a JSON object in the system prompt, or use `tokenizer.apply_chat_template()` with `tools=....`
2. **Function call:** By default, LFM2.5 writes Pythonic function calls (a Python list between `<|tool_call_start|>` and `<|tool_call_end|>` special tokens), as the assistant answer. You can override this behavior by asking the model to output JSON function calls in the system prompt.
3. **Function execution:** Execute the call and return the result with the `tool` role.
4. **Final answer:** LFM2.5 interprets the tool output and returns a plain-text answer addressing the original prompt.

See the [Tool Use documentation](#) for the full guide.

Example:

```
<|startoftext|><|im_start|>system
List of tools: [{"name": "get_candidate_statu
<|im_start|>user
```

```
What is the current status of candidate ID 1?
<|im_start|>assistant
<|tool_call_start|>[get_candidate_status(candidate_id=1)]
<|im_start|>tool
[{"candidate_id": "12345", "status": "Interviewed"}]
<|im_start|>assistant
The candidate with ID 12345 is currently in the interview stage.
```

 Inference

LFM2.5 is supported by many inference frameworks. See the [Inference documentation](#) for the full list.

Name	Description	Docs	Notebook
Transformers	Simple inference with direct access to model internals.	Link	 Open in Colab
vLLM	High-throughput production deployments with GPU.	Link	 Open in Colab
llama.cpp	Cross-platform inference with CPU offloading.	Link	 Open in Colab
MLX	Apple's machine learning framework optimized for Apple Silicon.	Link	—
LM Studio	Desktop application for running LLMs locally.	Link	—
SGLang	High-throughput production	Link	-

Name	Description	Docs	Notebook
	deployments with GPU.		

Quick start with Transformers (compatible with
transformers>=5.0.0):

```
from transformers import AutoModelForCausalLM

model_id = "LiquidAI/LFM2.5-230M"
model = AutoModelForCausalLM.from_pretrained(
    model_id,
    device_map="auto",
    dtype="bfloat16",
    # attn_implementation="flash_attention_2"
)

tokenizer = AutoTokenizer.from_pretrained(model_id)
streamer = TextStreamer(tokenizer, skip_prompt=True)

prompt = "What is C. elegans?"

input_ids = tokenizer.apply_chat_template(
    [{"role": "user", "content": prompt}],
    add_generation_prompt=True,
    return_tensors="pt",
    tokenize=True,
)["input_ids"].to(model.device)

output = model.generate(
    input_ids,
    do_sample=True,
    temperature=0.1,
    top_k=50,
    repetition_penalty=1.05,
    max_new_tokens=512,
    streamer=streamer,
)
```

Fine-Tuning

We recommend fine-tuning LFM2.5 for your specific use case to achieve the best results.

Name	Description	Docs	Notebook
CPT (Unsloth)	Continued Pre-Training using Unsloth for text completion.	Link	 Open in Colab
CPT (Unsloth)	Continued Pre-Training using Unsloth for translation.	Link	 Open in Colab
SFT (Unsloth)	Supervised Fine-Tuning with LoRA using Unsloth.	Link	 Open in Colab
SFT (TRL)	Supervised Fine-Tuning with LoRA using TRL.	Link	 Open in Colab
DPO (TRL)	Direct Preference Optimization with LoRA using TRL.	Link	 Open in Colab
GRPO (Unsloth)	GRPO with LoRA using Unsloth.	Link	 Open in Colab
GRPO (TRL)	GRPO with LoRA using TRL.	Link	 Open in Colab

Performance

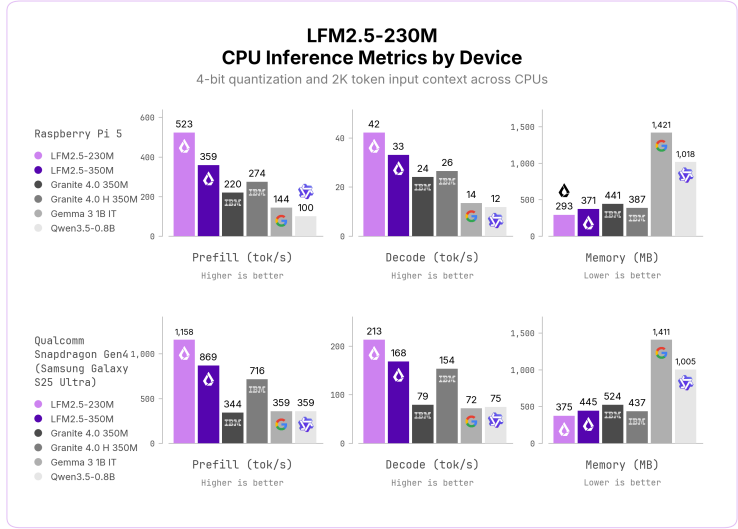
Benchmarks

Model	GPQA Diamond	MMLU- Pro	IFEval	IFBench	Multi- IF
LFM2.5- 230M	25.41	20.25	71.71	38.40	37.70
LFM2.5- 350M	30.64	20.01	76.96	40.69	44.90
LFM2- 350M	27.58	19.29	64.96	18.20	32.90
Granite 4.0-H- 350M	22.32	13.14	61.27	17.22	28.70
Granite 4.0-350M	25.91	12.84	53.48	15.98	24.20
Qwen3.5- 0.8B (Instruct)	27.41	37.42	59.94	22.87	41.60
Gemma 3 1B IT	23.89	14.04	63.49	20.33	44.20

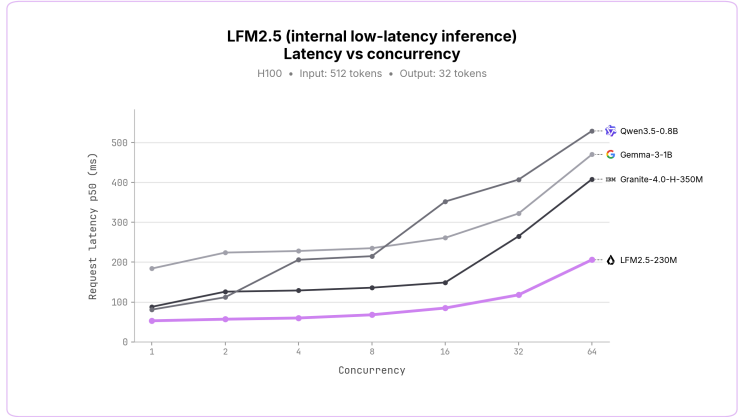
Model	CaseReport Bench	BFCLv3	BFCLv4	Task Be Tele
LFM2.5- 230M	22.51	43.26	21.03	5.26
LFM2.5- 350M	32.45	44.11	21.86	18.80
LFM2- 350M	11.67	22.95	12.29	10.80
Granite 4.0-H-	12.44	43.07	13.28	13.70

Model	CaseReport Bench	BFCLv3	BFCLv4	Be Tele
350M				
Granite 4.0-350M	0.84	39.58	13.73	2.92
Qwen3.5-0.8B (Instruct)	13.83	35.08	18.70	12.5
Gemma 3 1B IT	2.28	16.61	7.17	9.36

CPU Inference



GPU Inference



Contact

- Got questions or want to connect? [Join our Discord community](#).
- If you are interested in custom solutions with edge deployment, please contact [our sales team](#).

Citation

```
@article{liquidAI2026230M,  
  author = {Liquid AI},  
  title = {LFM2.5-230M: Built to Run Anywhere},  
  journal = {Liquid AI Blog},  
  year = {2026},  
  note = {www.liquid.ai/blog/lfm2-5-230m},  
}
```

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@article{liquidai2025lfm2,
```

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  title = {Technical Report},  
  author = {Liquid AI},  
  journal = {arXiv preprint arXiv:2511.23404},  
  year = {2025}  
}
```

 System theme

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